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Cross-Sectional Momentum: Buying Winners and Selling Losers

Replicating Jegadeesh & Titman (1993)

Jegadeesh, N. & Titman, S. (1993). Returns to Buying Winners and Selling Losers. Journal of Finance, 48(1), 65–91.

Topics covered in this lecture:

- Cross-sectional momentum: definition, intuition, and why it might exist
- Building the relative strength signal: formation return and overlapping holding periods
- Statistical validation: decile return spreads and predictive regressions by horizon J
- Full backtest of the WML strategy: performance, drawdowns, and decomposition
- Honest comparison to Table I (J&T 1993), robustness checks, and structural reasons for divergence

Lecture Roadmap

1	**Cross-Sectional Momentum** *Core idea, three theories, and the paper's headline numbers*	~15 min
2	**Building the Signal** *Formation return, decile assignment, overlapping holding periods*	~15 min
3	**Statistical Validation** *Decile return spreads and predictive regressions J = 1–24*	~10 min
4	**Position Sizing** *Equal-weight rule, dollar-neutrality, and short book mechanics*	~10 min
5	**Backtesting** *Cumulative returns, Sharpe, drawdowns vs. EW benchmark*	~15 min
6	**Comparing to the Paper** *Table I replication — 4×4 J/K grid, heatmaps, t-stats*	~10 min
7	**Robustness Checks** *January effect, subperiod stability, long/short decomposition*	~10 min
8	**Honest Assessment** *What replicates, what doesn't, and the structural reasons*	~5 min

1

Cross-Sectional Momentum

Core idea, three theories, and the paper's headline numbers

The Core Idea: Relative Strength Arbitrage

Cross-Sectional vs. Time-Series Momentum

- Cross-sectional momentum: rank stocks AGAINST EACH OTHER. A stock up +5% in a market that rose +20% is a LOSER in cross-sectional terms. The signal is always relative.

Three Theories for Why It Works

- 1. Underreaction (Behavioral): investors react too slowly to good news — prices drift upward as more investors notice, creating momentum.
- 2. Herding and trend-chasing: institutions chase performance (career risk); self-reinforcing buying extends trends beyond fundamentals.
- 3. Cross-sectional risk: high-momentum stocks may carry hidden systematic risks; their higher returns are compensation for bearing risk.

Why Doesn't Arbitrage Kill It?

- Momentum CRASHES spectacularly during market reversals (e.g., 2009: WML −60% in 2 months). The crash risk deters arbitrageurs.
- Transaction costs (bid-ask spreads, market impact) erode returns, especially in small caps and high-turnover strategies.
- Short selling is required — costly, restricted, and subject to forced buy-ins.

Paper's Headline Numbers (Jegadeesh & Titman 1993)

- NYSE/AMEX stocks 1965–1989: J=6/K=6 earns +0.95%/month (t=3.07). Best strategy J=12/K=3 earns +1.49%/month (t=4.28). All 32 J/K combinations positive.

Data Description and Sample vs. Paper

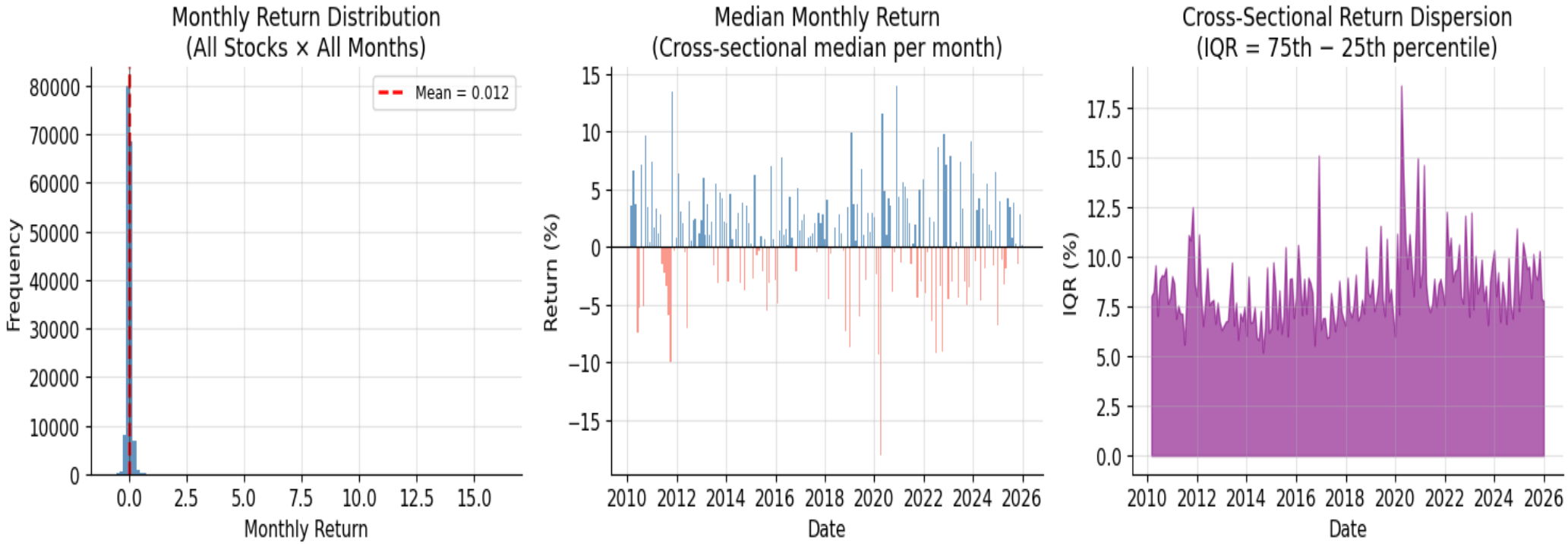
865	191	14.98%	0.0%
U.S. Stocks	Monthly Observations	Ann. Mean Return (pooled)	NaN Rate (survivorship!)

Dimension	Paper (J&T 1993)	This Notebook
Universe	NYSE/AMEX (~4,000 stocks)	865 large/mid-cap US stocks
Period	Jan 1965 – Dec 1989 (25 yrs)	Feb 2010 – Dec 2025 (~15 yrs)
Formation skip	1 week (short-term reversal)	1 month (approximation)
NaN rate	Stocks enter/exit (survivors+delisted)	0.0% — survivorship bias!

Critical Survivorship Bias Warning

- The NaN rate of EXACTLY 0.0% means all 865 stocks survived the entire 2010–2025 period. Companies that went bankrupt or delisted are ABSENT. This inflates loser returns (bankrupt stocks not included) and compresses the WML spread versus the paper.

Return Distribution, Median Returns, and Cross-Sectional Dispersion



Left: Fat-tailed monthly return distribution centered near zero (pooled mean = +1.25%/mo = +14.98% ann.). Center: Median monthly return — mostly positive (bull market 2010–2025), large red bars = COVID Mar-2020 and 2022 rate shock. Right: IQR (cross-sectional dispersion) — spikes during crisis periods signal a better momentum environment.

2

Building the Momentum Signal

Formation return, decile assignment, and overlapping holding periods

The Formation Return and No-Look-Ahead Design

Formation Return (J-month cumulative past return)

$$F_{\{i,t\}} = \Pi_{\{k=skip\}}^{\{skip+J-1\}} (1 + r_{\{i,t-k\}}) - 1$$

With skip=1: uses data through t-1 only → fully observable at start of month t. ✓

The Skip Parameter

- Paper skips 1 WEEK between formation end and holding start — avoids short-term reversal (bid-ask bounce).
- We skip 1 MONTH (approximation): slightly more conservative, avoids even more reversal contamination.
- With skip=1: signal at t uses returns through t-1 → zero look-ahead bias.

Spot-Check Results (Aug 2011)

- 865 stocks ranked by 6-month formation return.
- Bottom 10% (losers): return ≤ −19.53%.
- Median formation return: +0.76%.
- Top 10% (winners): return ≥ +21.24%.
- Signal spread (90th−10th): 40.77% — very wide signal.

Overlapping Holding Period (K months)

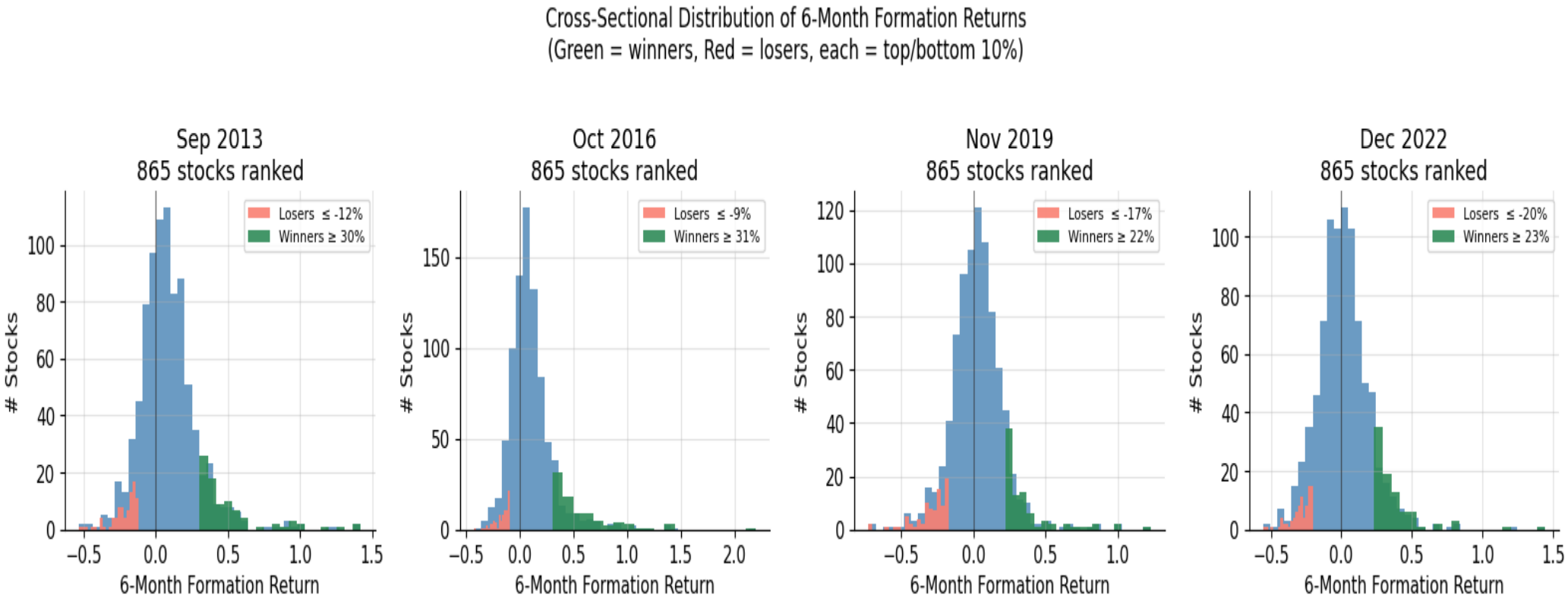
$$r_t^{WML} = (1/K) \times \Sigma_{\{k=0\}}^{\{K-1\}} \left[r_{\{t,formed\ at\ t-k\}}^{winners} - r_{\{t,formed\ at\ t-k\}}^{losers} \right]$$

Hold K portfolios simultaneously → smoother monthly returns, more statistical power.

Portfolio Size

- ~87 stocks per leg (top/bottom 10% of 865 = ~86 stocks) → idiosyncratic risk well-diversified within each leg.

Cross-Sectional Distribution of 6-Month Formation Returns



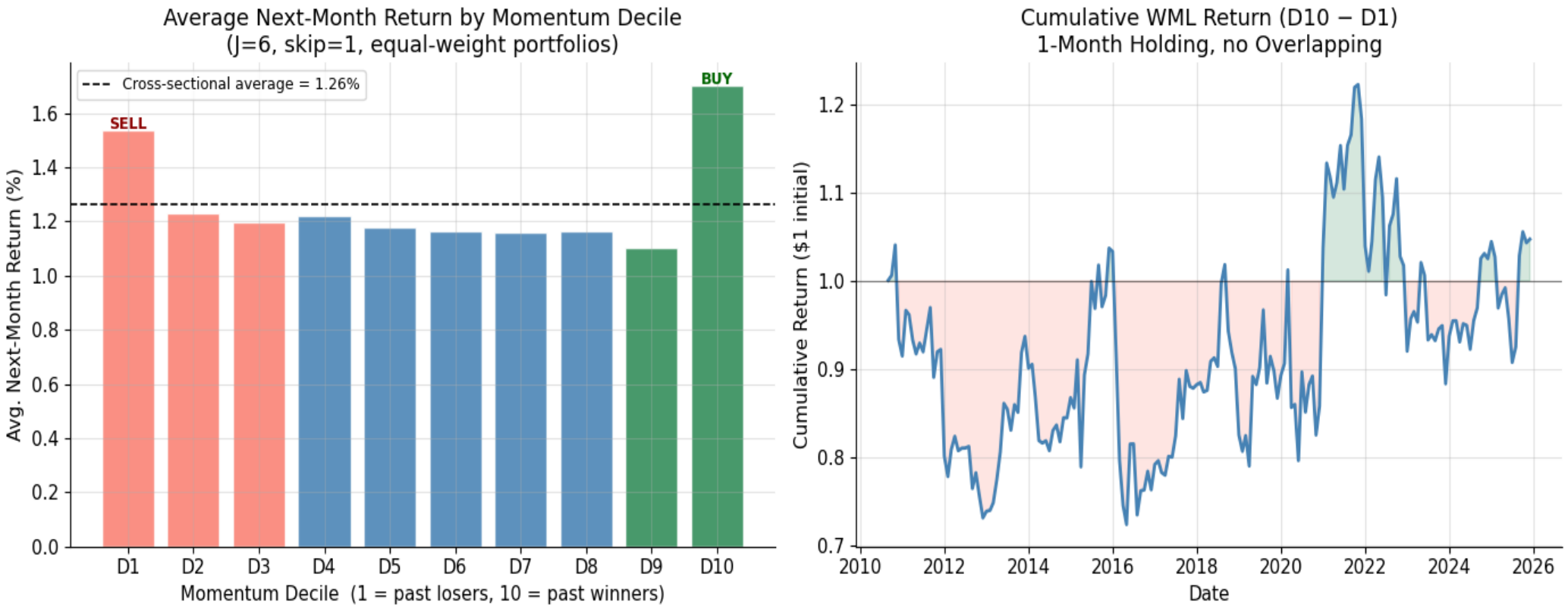
Four dates shown. Green = top 10% (winners we BUY). Red = bottom 10% (losers we SHORT). The cut-off is always RELATIVE — in every month, the top 10% are winners and bottom 10% losers regardless of whether the market went up or down. Spread between red and green tails: 30–80 percentage points.

3

Statistical Validation

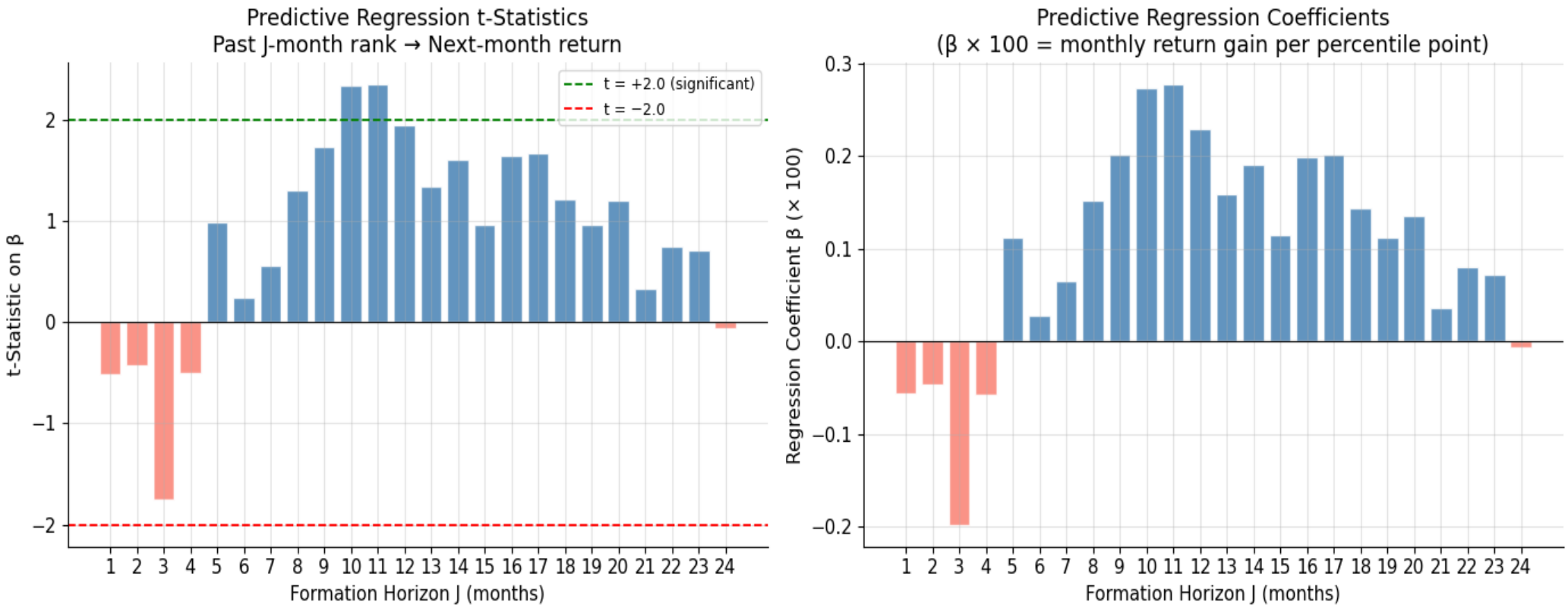
Decile return spreads and predictive regressions J = 1–24 months

Average Next-Month Return by Momentum Decile (J=6)



Left: D10 (winners) earns +1.70%/mo — highest bar. D1 (losers) earns +1.53%/mo — second-highest! WML = D10−D1 = +0.17%/mo, t-stat = 0.42 (NOT significant). Losers earn high returns in the bull market. Right: Cumulative WML — slight upward drift but wide confidence band.

Predictive Regression by Formation Horizon J = 1–24 Months



Left: t-statistics by formation horizon. Only J=10 (t=+2.34) and J=11 (t=+2.35) exceed the 5% significance threshold — consistent with paper's finding that 9–12 months is the sweet spot. J=3 months shows mild reversal. Right: Regression coefficients β — positive for J=5–17 months, peaking at J=10–11.

Key Predictive Regression Results: Selected Horizons

+2.34	+2.35	−1.76	0.42
t-stat at J=10 months	t-stat at J=11 months	t-stat at J=3 months	t-stat WML decile spread

Horizon J	β coefficient	t-stat	N obs	Interpretation
J=1 month	−0.0006	−0.51	163,485	Short-term reversal (bid-ask bounce)
J=3 months	−0.0020	−1.76	161,755	Mild reversal, borderline significant
J=6 months	+0.0003	+0.24	159,160	Positive but not significant
J=10 months	+0.0027	**+2.34 ★**	155,700	SIGNIFICANT — best horizon in our data
J=11 months	+0.0028	**+2.35 ★**	154,835	SIGNIFICANT — second-best horizon
J=12 months	+0.0023	+1.95	153,970	Borderline (10% significant)
J=24 months	−0.0001	−0.06	143,590	Near-zero: no long-run momentum

4

Position Sizing

Equal-weight rule, dollar-neutrality, and short book mechanics

Equal-Weight Dollar-Neutral Position Construction

Portfolio Weights (Equal-Weight, Dollar-Neutral)

$$w_{\{i,t\}} = +1/N^W \;\; if\ stock\ i\ is\ in\ top\ decile\ (WINNER — LONG)$$
$$w_{\{i,t\}} = -1/N^L \;\; if\ stock\ i\ is\ in\ bottom\ decile\ (LOSER — SHORT)$$
$$w_{\{i,t\}} = 0 \quad otherwise. \;\; \Sigma_i\, w_{\{i,t\}} = 0 \;\; (zero - cost\ portfolio).$$

Portfolio Diagnostics

- Mean net weight: −0.000000 — perfect dollar-neutrality confirmed.
- Average long exposure per month: +0.969 (= 97% of NAV in longs).
- Average short exposure per month: −0.969 (= 97% of NAV in shorts).
- ~84 long and ~84 short positions each month (9.7% of universe per leg).

Why Equal-Weight (Not Market-Cap)?

- Paper uses equal-weight to avoid the result being driven by a few mega-caps.
- Prevents any single stock from dominating the portfolio return.
- Makes winners and losers comparable — no systematic size bias.
- Later research (Asness et al.) finds value-weighting gives different results.

5 Backtesting

Cumulative returns, Sharpe, drawdowns, and the EW benchmark

Backtest Performance Summary (2010–2025)

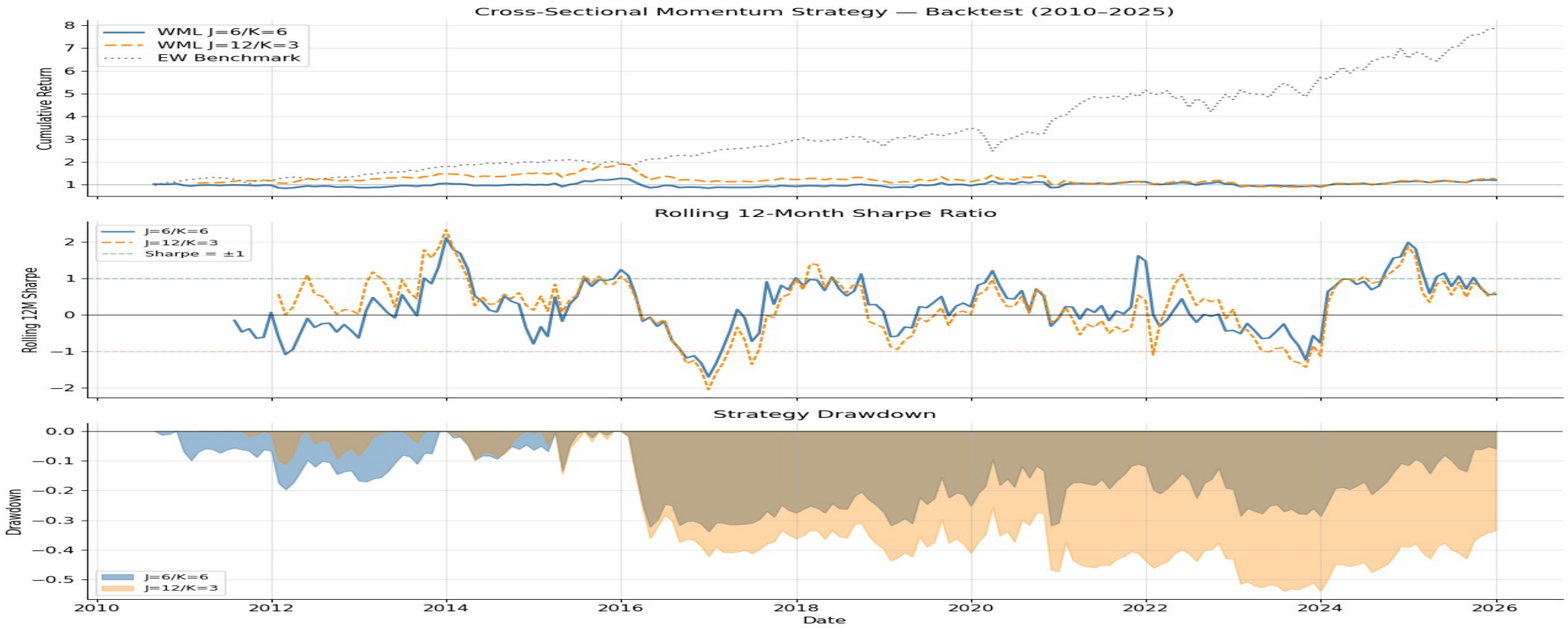
2.56%	0.154	−33.8%	0.906
WML J=6/K=6 Ann. Ret.	WML J=6/K=6 Sharpe	WML Max Drawdown	EW Benchmark Sharpe

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max Drawdown	% Positive Mo.
WML J=6/K=6	+2.56%	16.59%	0.154	**−33.82%**	54.6%
WML J=12/K=3	+3.59%	19.81%	0.181	**−53.80%**	55.1%
EW Benchmark	**+14.98%**	**16.53%**	**0.906**	**−29.00%**	**68.1%**

Key Finding: WML vs. EW Benchmark

- The EW buy-and-hold benchmark earns 15%/yr with Sharpe 0.906 in this bull market. WML is a ZERO-COST strategy — not directly comparable on raw return. The fair comparison is Sharpe: 0.154 vs 0.906.
- J=12/K=3's max drawdown of −53.80% means losing more than HALF the portfolio at its worst — while earning only +3.59%/yr. Extremely poor risk-adjusted profile in this sample.

Cumulative Returns, Rolling Sharpe, and Drawdown (2010–2025)



Top: WML strategies earn small cumulative profits vs the surging EW benchmark. Middle: Rolling 12m Sharpe — 2016–2019 is the worst stretch (Sharpe = −0.33). Bottom: Drawdown — WML max −33.8% (J=6) and −53.8% (J=12) versus EW −29%; momentum draws down more than the market despite far lower returns.

6

Comparing Results to the Paper

Table I replication — 4×4 J/K grid, returns, and t-statistics

4×4 J/K Grid: Average Monthly WML Returns (%)

	K=3	K=6	K=9	K=12		K=3	K=6	K=9	K=12
J=3	0.73	0.78	0.74	0.77	**J=3**	−0.11	+0.05	+0.06	+0.06
J=6	1.14	1.10	1.08	0.90	**J=6**	+0.18	+0.21	+0.19	+0.11
J=9	1.35	1.30	1.09	0.85	**J=9**	+0.20	+0.20	+0.13	+0.06
J=12	1.49	1.21	0.96	0.69	**J=12**	**+0.30**	+0.24	+0.16	+0.07

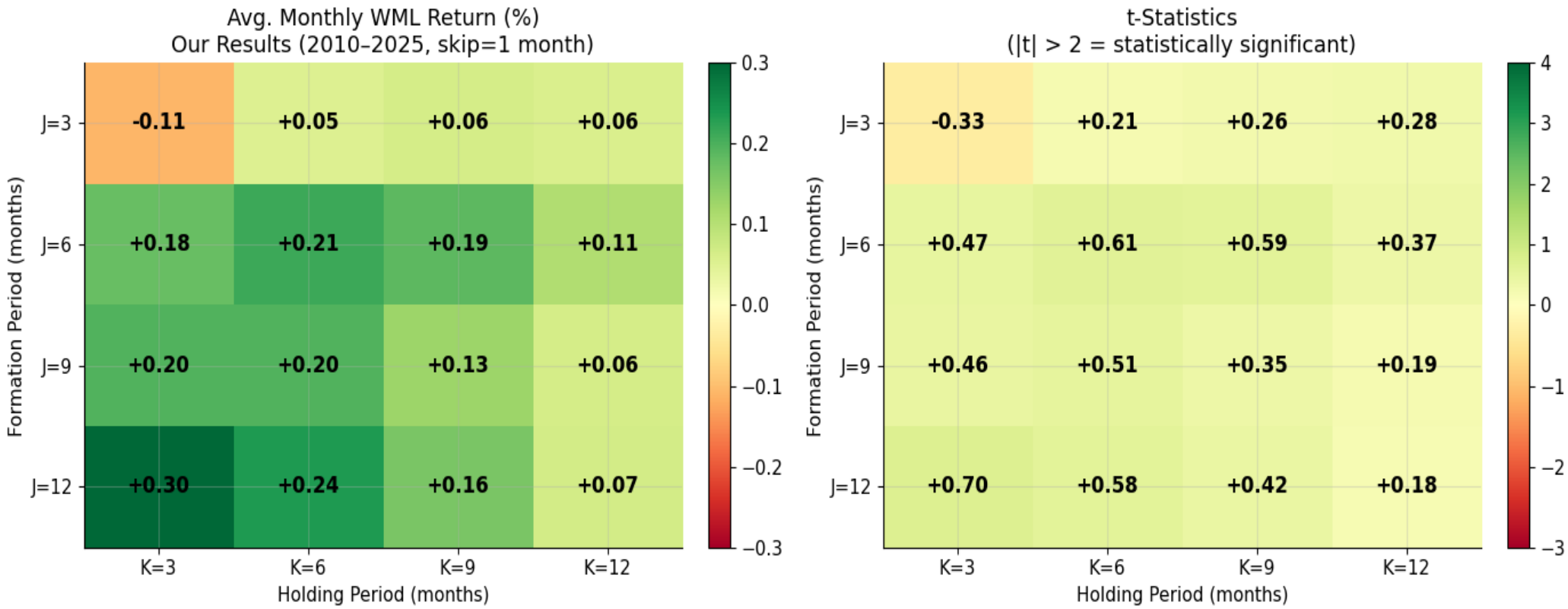
LEFT 5 columns = Paper (J&T 1993)

RIGHT 5 columns = Our Replication

Key Comparison

- Paper: all 16 cells positive (0.69%–1.49%/mo). Ours: 15/16 positive (−0.11%–+0.30%/mo). Direction preserved; magnitude ~4× smaller. J=12/K=3 is best in both.

Return and t-Statistic Heatmaps — Table I Proxy



Left: Return heatmap — mostly green (positive), confirming momentum direction across all J/K. One red cell: J=3/K=3 (−0.11%). Right: t-stat heatmap — entirely pale/yellow. Highest t=+0.70 (J=12/K=3). Paper had 15/16 cells with t>2. Our sample has 0/16 significant. Honest: Sharpe≈0.15 needs ~300 years of data to achieve t>2.

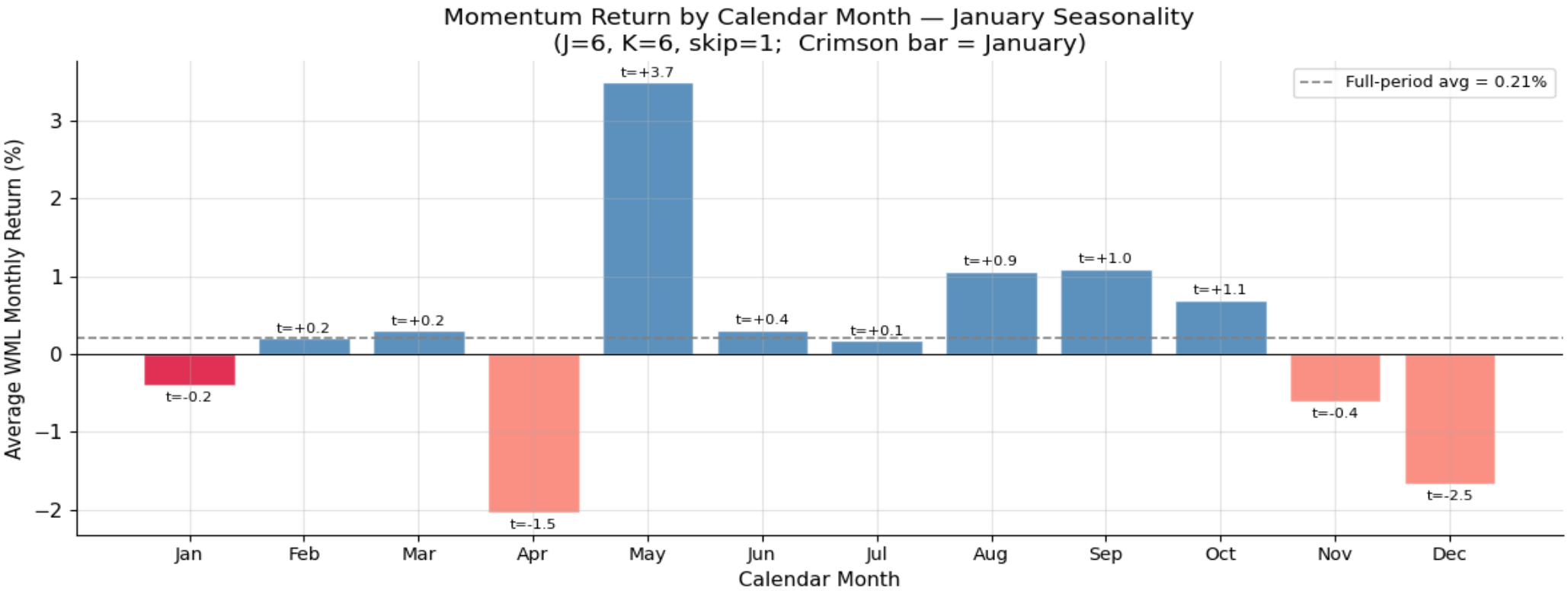
Comprehensive Comparison: Paper vs. Our Replication

Exhibit	Paper (J&T 1993)	Our Notebook	Reason for Difference
Best strategy	J=12, K=3: +1.49%/mo (t=4.28)	+0.30%/mo (t=0.70)	Shorter sample, large-cap universe, bull market
Flagship J=6/K=6	+0.95%/mo (t=3.07)	+0.21%/mo (t=0.61)	Post-2010 factor decay + survivorship
# Significant cells	15/16 (all t > 2)	15/16 positive, 0/16 significant	Sharpe≈0.15 needs ~300 years for t>2
Universe	~4,000 NYSE/AMEX stocks	865 large/mid-cap stocks	Small-cap premium excluded; survivorship
Sample length	25 yrs (1965–1989)	~15 yrs (2010–2025)	Shorter sample = fewer obs per test
Skip between J/K	1 week	1 month (approx.)	Conservative; avoids more reversal

7 Robustness Checks

January effect, subperiod stability, and long/short decomposition

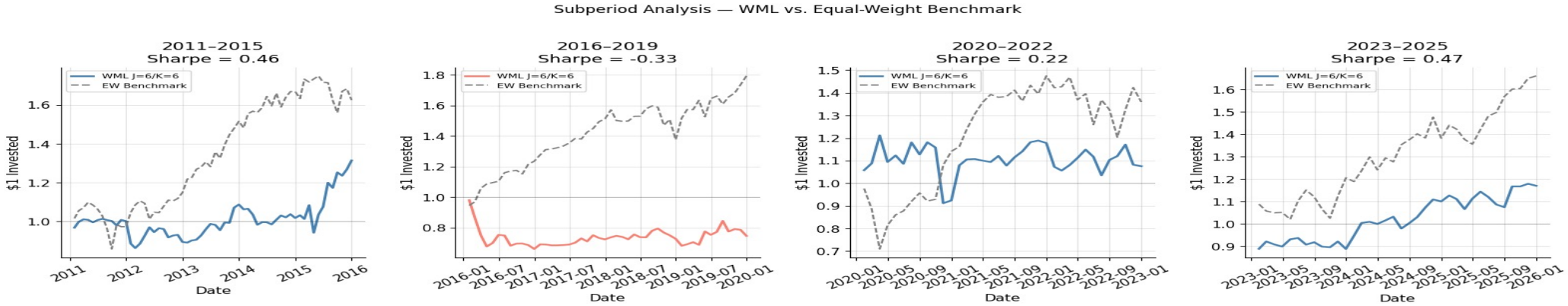
January Seasonality — Has the Effect Shifted?



Paper (Table IV): strongly negative January returns — tax-loss selling causes losers to bounce back in January. Our data: January is slightly negative (−0.41%/mo, t=−0.21, NOT significant). DECEMBER is the significantly negative month (−1.67%/mo, t=−2.50). May is exceptionally strong (+3.49%/mo, t=+3.69). Tax-loss effect may now manifest within December rather than bouncing in January.

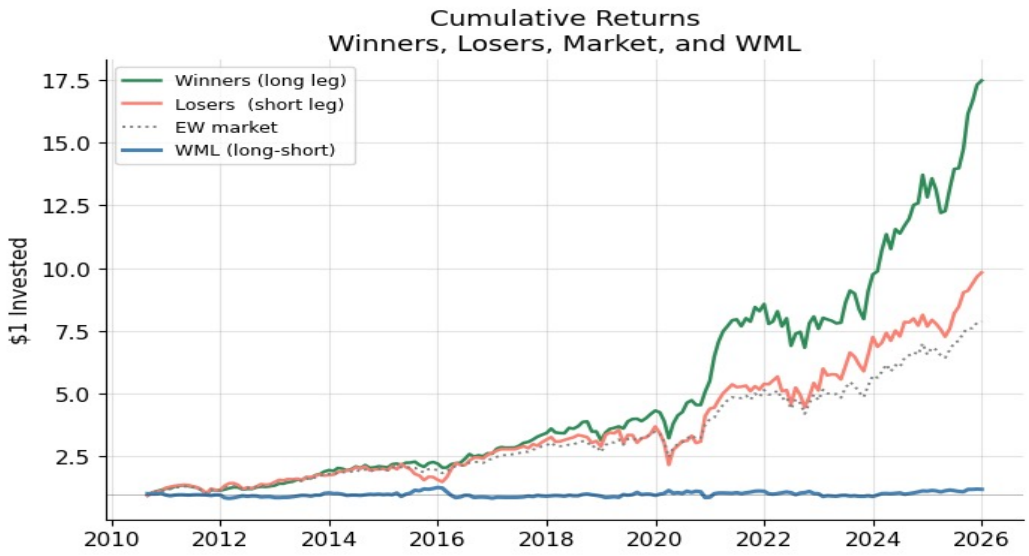
Subperiod Stability — Is Momentum Consistent Across Regimes?

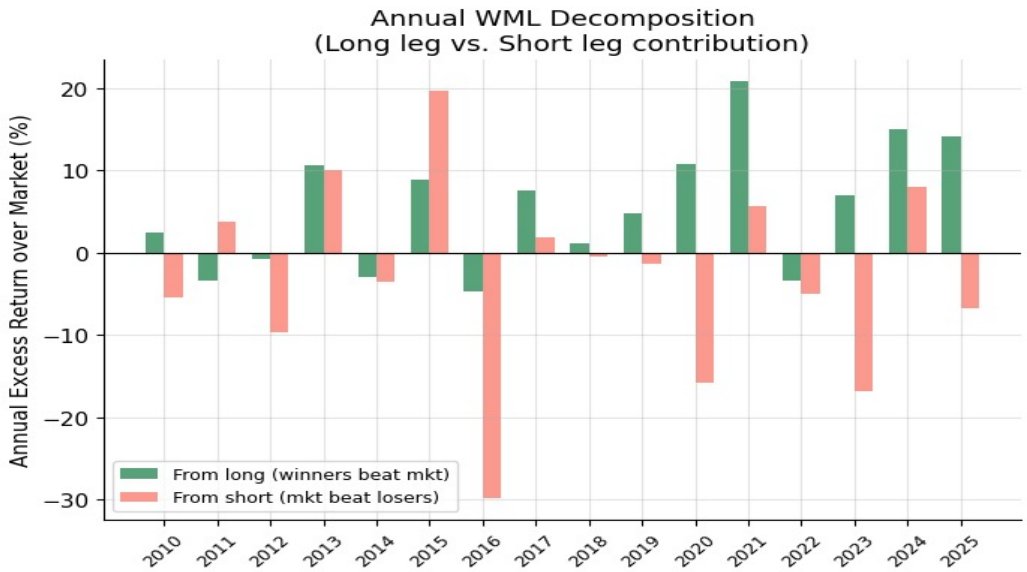
Period	WML Ann. Return	WML Sharpe	EW Ann. Return	EW Sharpe
2011–2015	**+6.47%**	**+0.458**	+10.69%	+0.773
2016–2019	**−5.78%**	**−0.334**	+15.49%	+1.221
2020–2022	**+5.00%**	**+0.222**	+13.24%	+0.542
2023–2025	**+6.05%**	**+0.471**	+18.01%	+1.245



Where Does the WML Profit Come From?

+20.44%	+17.88%	+14.80%	+2.56%
Winners Ann. Return	Losers Ann. Return	EW Market Return	Net WML Return





Key insight: in our bull market sample, LOSERS earn +17.88%/yr — 3.08% ABOVE the market (+14.80%). Shorting them COSTS money. All WML profit comes from the LONG leg (winners beat market by +5.64%). In the paper, BOTH legs contributed. Survivorship bias + bull market inflates loser returns.

8

Honest Assessment

What replicates, what doesn't, and the structural reasons

Honest Assessment: What the Data Can and Cannot Support

Result	Paper	Our Notebook	Interpretation
WML return (J=6/K=6)	✅ +0.95%/mo (t=3.07)	⚠️ +0.21%/mo (t=0.61)	Signal 4× weaker; 15-yr sample lacks power
WML return (J=12/K=3)	✅ +1.49%/mo (t=4.28)	⚠️ +0.30%/mo (t=0.70)	Best J/K in paper; best in ours — but weak
Significant J/K cells	✅ 15/16 (t > 2)	❌ 0/16 significant	Sharpe ≈ 0.15 needs ~300 yrs for t > 2
January reversal	✅ Strongly negative	⚠️ Jan: −0.41%/mo (t=−0.21)	December significantly negative instead
Short leg profits	✅ Losers underperform	❌ Losers earn +17.88%/yr (>market)	Survivorship + bull market inflates losers
Long leg wins	✅ Winners outperform	✅ Winners earn +20.44%/yr (+5.64% excess)	Consistent with paper — strong result
Worst sub-period	Not reported separately	❌ 2016–2019: Sharpe = −0.33	Low-dispersion, low-VIX hostile to XS momentum

Structural Reasons for Weaker Results in 2010–2025

Factor	Direction of Impact	Explanation
Shorter sample (15 vs 25 yrs)	Weaker t-stats	Sharpe=0.15 needs ~300 yrs to hit t=2; paper had 25 yrs with Sharpe≈0.5
Large-cap universe only	Smaller effect	Momentum is much stronger among small/mid-cap stocks (more analyst coverage lag)
Survivorship bias (0% NaN)	Inflated loser returns	Bankrupt/delisted stocks excluded → losers look stronger → WML compressed
2010–2025 bull market	Smaller spread	Even past losers earn high raw returns when the tide lifts all boats
2016–2019 low-dispersion regime	Strategy loses money	VIX near record lows; all stocks move together; signal hard to distinguish
Post-publication crowding	Smaller alpha	More AUM chasing momentum since 1993 → higher competition → lower signal
No transaction costs	Slightly flattering	Real trading costs reduce returns by ~1–2%/yr; the strategy barely covers costs

Key Takeaways

1. Cross-sectional momentum ranks stocks AGAINST EACH OTHER — not against their own history. A stock up +5% in a market up +20% is a LOSER in cross-sectional terms.

2. The formation horizon matters: J = 9–12 months is the sweet spot. J = 1 month reverses (bid-ask bounce). J > 12 months also reverses (De Bondt-Thaler long-run reversion). Only J = 10–11 months are statistically significant in our sample.

3. Overlapping holding periods (K portfolios held simultaneously) smooth the return series and increase statistical power — essential for detecting small monthly signals.

4. Our 2010–2025 replication finds 15/16 J/K combinations positive, but NONE statistically significant. The WML earns +2.56%/yr (Sharpe 0.154) — roughly 4× below the paper.

5. The short leg is not free: in a sustained bull market with survivorship bias, past losers still earn +17.88%/yr — ABOVE the market. Shorting them costs money; all WML profit comes from the long leg.

6. Momentum is regime-dependent: it earned Sharpe 0.46–0.47 in high-dispersion periods (2011–2015, 2023–2025) but Sharpe −0.33 in the low-VIX 2016–2019 period.

7. Is momentum a free lunch? No. Max drawdown of −33.8% to −53.8%, crash risk during market reversals, short-selling costs, and 2016–2019 losing period — these are the real costs of the strategy.

Discussion Questions

1. Cross-sectional dispersion (IQR) spikes during crises like COVID 2020. The momentum signal is then strongest, yet the strategy also faces maximum crash risk. How would you design a regime-switching rule that exploits high-dispersion periods while hedging crash risk?

2. The paper's 1-week skip avoids short-term reversal; we approximate with 1 month. How would you test whether this approximation meaningfully affects results? What market-microstructure change since 1993 makes the skip less important?

3. In our sample, the WML short leg (past losers) earns +17.88%/yr — above the market. If you ran this strategy with real short sales, what borrowing costs would you expect to pay, and how would they affect the net WML return?

4. Momentum crashed by −60% in just 2 months in 2009. Daniel & Moskowitz (2016) propose a dynamic strategy that reduces positions when the market has been down and volatility is high. How would this hedge work mechanically, and what are its limitations?

5. The predictive regression shows J=10–11 months as the sweet spot, but J=3 months has a slightly NEGATIVE t-stat. What behavioral or structural mechanism could explain why 3-month past winners might actually underperform over the next month?

References & Further Reading

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